Comparison Between Force and Ultrasound Image-Based Controllers for Autonomous Robotic Ultrasound Acquisition in Different Tissue Types

F. J. Domingo Gil, A. G. de Groot(✉), and F. J. Siepel

Robotics and Mechatronics Group, University of Twente, Enschede, The Netherlands
a.g.degroot@utwente.nl

Abstract. The main challenges in automatic robotic ultrasound (US) scanning include ensuring acoustic coupling between the tissue and the US probe, while minimizing tissue deformation. In this comparative study, a force controller, confidence-driven controller (CDC), and hybrid force-image controller are tested to perform automatic robotic ultrasound (US) scan. The controllers have been tested using a 7DoF robot equipped with a force sensor and a linear US transducer. Obtaining a robotic 3D US acquisition would increase the reproducibility and accuracy of the diagnostic, and at the same time, would simplify the registration of 3D US volumes with MRI images. The robot follows a reference trajectory which is adjusted by any of the controllers to ensure acoustic coupling. The CDC makes the corrections based on confidence maps. The force and hybrid controllers correct the probe contact based on the contact force of the end-effector with the tissue. The CDC and hybrid controller ensure normal orientation to the tissue by controlling in-plane rotations based on the confidence map. The three controllers are tested in three phantoms of different stiffness.

Keywords: autonomous US acquisition · force controller · confidence maps

1 Introduction

Breast cancer remains a global health concern, as 1 out of 8 women are diagnosed with breast cancer in their life. Early detection through reliable imaging modalities such as mammography, MRI, and ultrasound (US) is essential for positive prospects [5]. While US-guided biopsies are preferred for their real-time feedback, they face challenges like depth assessment in 2D scans, leading to potential misdiagnosis. The possibility of obtaining direct 3D volumes has been studied, by utilizing mechanical 3D probes or freehand scanners [2]. However, all these solutions depend on highly-trained human operators, what reduces the repeatability of the process. To address this challenge, the possibility of performing autonomous robotic US acquisition is studied.

C. Secchi and L. Marconi (Eds.): ERF 2024, SPAR 33, pp. 394–398, 2024.
https://doi.org/10.1007/978-3-031-76428-8_73

Robotic systems offer advantages in trajectory precision, reduced variability, and higher repeatability. Despite these benefits, challenges like patient movement or tissue deformation may affect US image quality during autonomous robotic US acquisition. To solve these issues, control strategies, including image feedback, force feedback, and their combination, have been developed. Image feedback, uses confidence maps, to ensure acoustic coupling and optimize image quality [1,4]. Force feedback maintains a constant force to minimize tissue deformation, while a hybrid approach combines force and image feedback for optimal control.

In this study, a force feedback controller and a hybrid force-image feedback controller for automated robotic US scanning are developed. Its behaviour is tested in three different phantoms with different stiffness, and compared to the behavior of a confidence map controller. Testing the controllers in different tissue types will provide insight into the optimal situation to use each of them.

2 Materials and Methods

2.1 Robotic Setup

For this research, a KUKA LBR MED 7 robotic arm was used. Attached to the flange of the KUKA arm is an ATI Mini-40 Force/Torque (F/T) sensor (Manufacturer page). Additionally, at the top of the sensor, an end-effector (EE) which includes a linear US transducer is fixed. Three phantoms are used to mimic different tissue types and complement the robotic setup. The phantoms were fabricated with a solution mixture of Plastisol and Softener in the three following concentration ratios: 100/0, 80/20, and 60/40 (Plastisol/Softener).

2.2 Control Strategy

The robot follows a predefined trajectory, which ensures that the whole tissue is scanned. However, trajectory errors and environmental uncertainties, like patient movements or breath, can affect acoustic coupling. Additionally, for optimal registration with modalities such as MRI, tissue deformation must be minimized. Three different controllers are implemented to address these challenges.

2.3 Confidence Driven Controller (CDC)

As Welleweerd et al. reported in their study, confidence maps can be used to control two degrees of freedom of the probe: the in-plane rotation and the translation in the z-direction of the EE frame, ψ_{ee} [4]. Please refer to [4] for a detailed explanation of this controller.

2.4 Force Controller

To use the force sensor data, a calibration procedure was developed. It ensures bias calibration, removal of the EE static mass [3], and reference frame alignment.

In the force control algorithm, F_z can be used to satisfy a constant contact force setpoint that ensures acoustic coupling, by controlling the z-translation in the EE reference frame. The error is determined by the difference between the force setpoint, F_{set}, and the current force reading in the z-direction, F_z: $e = F_{set} - F_z$. To control F_z a simple PD controller is used. The trajectory pose $H(i)^0_{ref}$ of the robot is adjusted by $H(i)^{ref}_{adj}$ at time i as follows:

$$H(i)^0_F = H(i)^0_{ref} H(i)^{ref}_{adj} = H(i)^0_{ref} H(i-1)^{ref}_{adj} H_{F\Delta z} \tag{1}$$

with

$$H_{F\Delta z} = \begin{bmatrix} I^{3\times 3} & d\hat{z} \\ 0^{1\times 3} & 1 \end{bmatrix}$$

$H(i)^{ref}_{adj}$ includes the current and previous outputs of the force control algorithm. The force control algorithm outputs a translation, Δz, in z-direction, $H_{F\Delta z}$.

2.5 Hybrid Force-Image Controller

An important factor in maximizing the quality of the US images is maintaining the US transducer normal to the scanning region. The force controller in the current setup lacks this capacity, therefore a hybrid force-image controller is used for normal orientation control. Similar to the previously explained force controller, the z-translation in the EE reference frame is controlled by F_z and a force setpoint. However, the in-plane rotation is controlled by the visual servoing algorithm, based on the confidence-weighted barycentre. Equation 1 is extended according to:

$$\begin{aligned} H(i)^0_{Fvs} = H(i)^0_{ref} H(i)^{ref}_{adj} = H(i)^0_{ref} H(i-1)^{ref}_{adj} H(i)^{adj}_{Fvs} \\ H(i)^{adj}_{Fvs} = H_{F\Delta z} H_{\Delta\theta} \end{aligned} \tag{2}$$

with

$$H_{F\Delta z} = \begin{bmatrix} I^{3\times 3} & d\hat{z} \\ 0^{1\times 3} & 1 \end{bmatrix} H_{\Delta\theta} = \begin{bmatrix} Rot_y(\theta) & 0^{3\times 1} \\ 0^{1\times 3} & 1 \end{bmatrix}$$

In addition to the z-translation obtained from the force control algorithm, the control algorithm outputs in this case a rotation $\Delta\theta$ around the y-axis of the transducer, $H_{\Delta\theta}$ as in the case of the CDC.

3 Results

Figure 1 shows a comparison of the mean confidence and its corresponding contact force, F_z, for each of the 10 experiments for each confidence setpoint in each phantom using the CDC. It can be seen that the confidence stays constant in the 3 phantoms, while the force varies. Furthermore, it is observed how F_z for 80% contact in the hard phantom differs a lot between tests.

To assess image quality and the effect of in-plane corrections, the mean barycenter error was used. Figure 2 shows the mean barycenter error of each

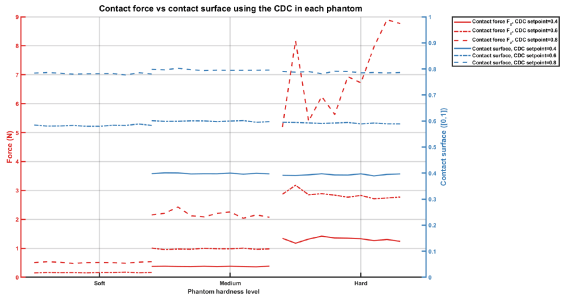

Fig. 1. Comparison of the contact force and contact surface using the CDC for the 3 setpoints in each phantom.

of the ten experiments performed with each controller for each setpoint. The CDC and hybrid controller both have in-plane normal control, while the force controller does not. The absence of in-plane correction significantly increases the barycenter error in all cases except the 80% hard, situation, which has similar values to the CDC and Hybrid controller.

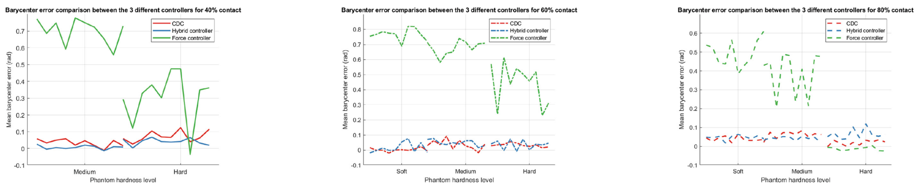

Fig. 2. Mean barycenter error for each of the 10 experiments for 40%, 60% and 80% contact in each phantom.

4 Discussion

In autonomous robotic US acquisition it is essential to ensure and optimize acoustic coupling and image quality while minimizing breast deformation. This comparative study shows three different controllers designed to satisfy the needs: a force controller, a CDC, and a hybrid force-image controller.

As observed in Fig. 1 the CDC is a more universal controller, since the setpoint to be used is independent of the phantom stiffness. On the contrary, the F_z obtained for each CDC setpoint varies depending on the phantom stiffness. These results show that the F_z needed to obtain a certain contact surface is highly dependent on the hardness of the phantom being scanned. Therefore, the force and hybrid controllers need individual setpoints for each phantom.

In Fig. 1 it is observed as well that, there is a lack of data for 40% confidence in the soft phantom. During the experiments, the CDC controller was tuned trying to simulate the results from Welleweerd et al. [4], therefore, the same controller was used for all the setpoints and phantoms. However, for 40% contact in the soft phantom, the controller was unstable and was losing contact with the tissue. An optimized tuning for each situation could make the experiments for 40% confidence in the soft phantom feasible, and could also improve all the results.

The results regarding in-plane rotation based on the barycenter of the confidence map also show great performance. While this is not a new feature and previous studies have demonstrated similar outcomes [4], this study has shown the potential of combining force feedback for z-translation control and confidence maps for in-plane rotation control. The barycenter error shows no significant difference between the CDC and Hybrid controller, while the barycenter error for the force controller, which does not control in-plane rotation is very high in most cases.

5 Conclusion

A force controller, a CDC, and a hybrid force-image controller for automatic robotic US acquisition were compared. The three controllers focus on optimizing acoustic coupling and image quality while minimizing breast deformation. The controllers were tested by scanning three different phantoms of three different stiffness. Results show that the CDC performs similarly in the three phantoms, applying a similar amount of deformation in any case using the same confidence setpoint. Therefore, the CDC has been proven to be independent of the stiffness, while the force and hybrid controllers are not. Considering that breast tissue can vary among individuals and does not have a uniform stiffness due to its complex anatomy, the CDC appears to be the most optimal controller for automatic robotic US scan of breast tissue. In addition, the results of the barycenter error show the importance of having in-plane rotation control to optimize the image quality.

References

1. Chatelain, P., Krupa, A., Navab, N.: Confidence-driven control of an ultrasound probe. IEEE Trans. Robot. **33**(6), 1410–1424 (2017). https://doi.org/10.1109/TRO.2017.2723618. ISSN 15523098
2. Huang, Q., Zeng, Z.: A review on real-time 3D ultrasound imaging technology. BioMed Res. Int. **2017** (2017). https://doi.org/10.1155/2017/6027029. ISSN 23146141
3. Vougioukas, S.: Bias estimation and gravity compensation for forcetorque sensors. In: Proceedings of International Conference on Mathematical Methods and Computational Techniques in Electrical Engineering, pp. 82–85. Citeseer (2001)
4. Welleweerd, M.K., et al.: Automated robotic breast ultrasound acquisition using ultrasound feedback. In: Proceedings - IEEE International Conference on Robotics and Automation, pp. 9946–9952 (2020). https://doi.org/10.1109/ICRA40945.2020.9196736. ISSN 10504729
5. Wilkinson, L., Gathani, T.: Understanding breast cancer as a global health concern 1. Br. J. Radiol. **95**, 20211033 (2021). https://doi.org/10.1259/bjr.20211033